

```
import pandas as pd
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score

# Data sesuai tabel soal
data = {
    "NIM": ["TI001", "TI002", "TI003", "TI004", "TI005", "TI006", "TI007", "TI008",
    "Actual": ["Lulus", "Lulus", "Lulus", "Lulus", "Lulus", "Tidak Lulus", "Tidak
    "Predicted": ["Lulus", "Lulus", "Lulus", "Tidak Lulus", "Tidak Lulus", "Lulus
}

df = pd.DataFrame(data)
df
```

	NIM	Actual	Predicted	
0	TI001	Lulus	Lulus	
1	TI002	Lulus	Lulus	
2	TI003	Lulus	Lulus	
3	TI004	Lulus	Tidak Lulus	
4	TI005	Lulus	Tidak Lulus	
5	TI006	Tidak Lulus	Lulus	
6	TI007	Tidak Lulus	Tidak Lulus	
7	TI008	Tidak Lulus	Tidak Lulus	
8	TI009	Tidak Lulus	Tidak Lulus	
9	TI010	Tidak Lulus	Tidak Lulus	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

Membuat Confusion Matrix

```
y_true = df["Actual"]
y_pred = df["Predicted"]

# Konversi label menjadi biner (Lulus = 1, Tidak Lulus = 0)
df_binary = df.replace({"Lulus": 1, "Tidak Lulus": 0}).infer_objects(copy=False)
y_true_bin = df_binary["Actual"]
y_pred_bin = df_binary["Predicted"]

cm = confusion_matrix(y_true_bin, y_pred_bin)
cm
```

```
/tmp/ipython-input-167357432.py:5: FutureWarning: Downcasting behavior in `re
df_binary = df.replace({"Lulus": 1, "Tidak Lulus": 0}).infer_objects(copy=False)
```

```
array([[4, 1],  
       [2, 3]])
```

Menghitung Accuracy, Precision, Recall

```
accuracy = accuracy_score(y_true_bin, y_pred_bin)  
precision = precision_score(y_true_bin, y_pred_bin)  
recall = recall_score(y_true_bin, y_pred_bin)  
  
accuracy, precision, recall
```

```
(0.7, 0.75, 0.6)
```

Kesimpulan

Berdasarkan hasil perhitungan, model prediksi kelulusan menghasilkan:

- Accuracy = 70% → Model cukup baik tetapi masih bisa ditingkatkan.
- Precision = 75% → Ketika model memprediksi "Lulus", 75% prediksi tersebut benar.
- Recall = 60% → Model hanya mampu menangkap 60% mahasiswa yang benar-benar lulus.